Cloud Security Auto-Remediation Architecture

Subtitle: Event-driven security automation system to detect and automatically remediate insecure cloud configurations in real time

Tech stack: Terraform · GitHub Actions · AWS CloudTrail · EventBridge · Lambda

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Role: Security Automation / Cloud Security

Keywords: CI/CD, IaC, Event-driven Security Automation

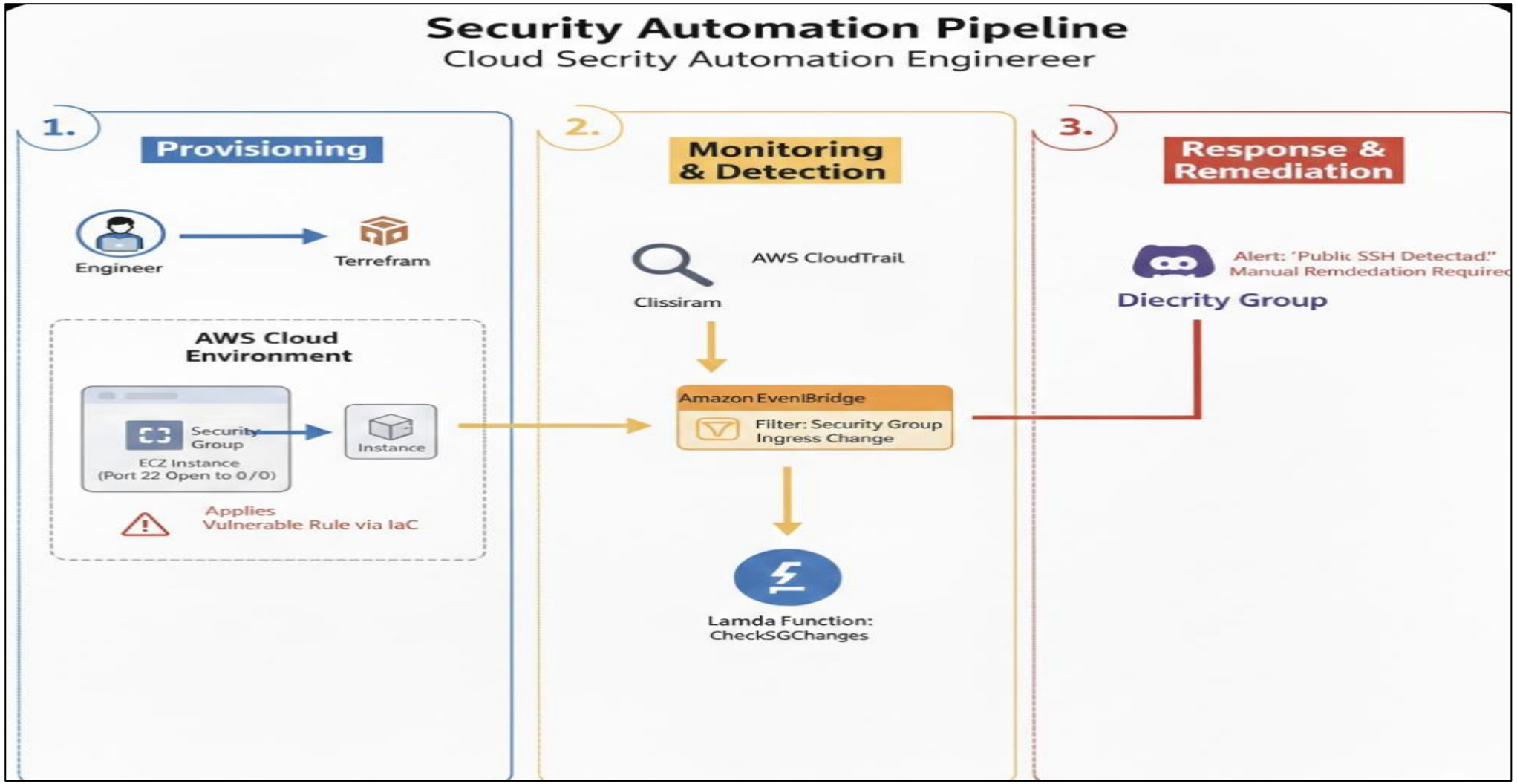
Problem Definition

- **Real-World Issue**

- In cloud environments, allowing SSH access from 0.0.0.0/0 in a Security Group
- is one of the most common initial attack vectors in security incidents.

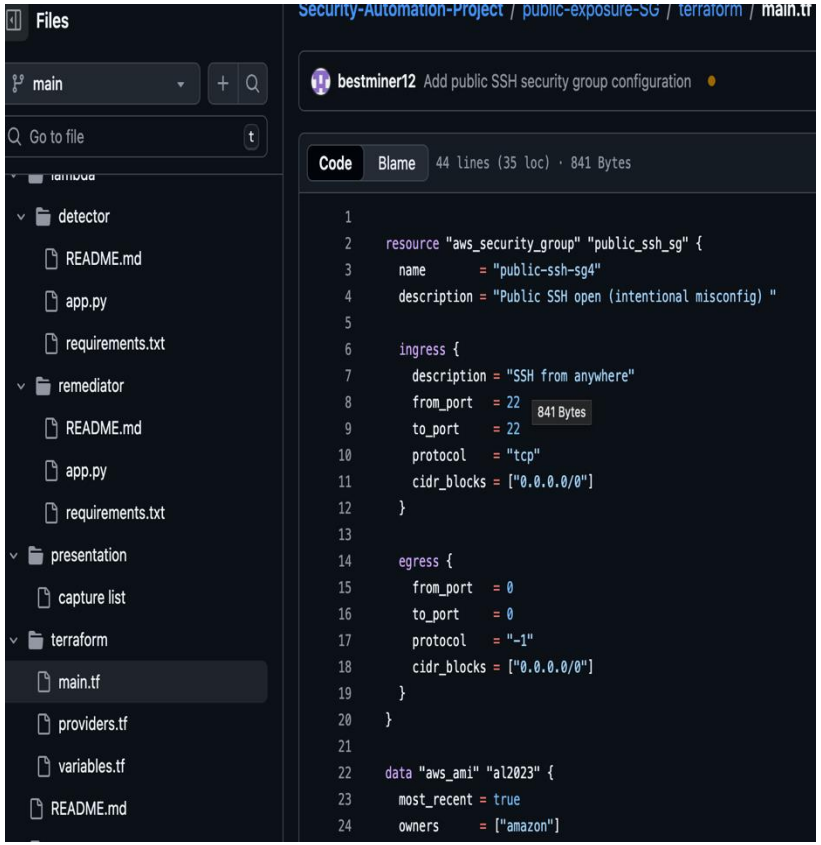
- In production environments:
- Manual detection → Delayed response
- Manual remediation → Human error
- Nights / Off-hours → Response gaps

High-Level Architecture

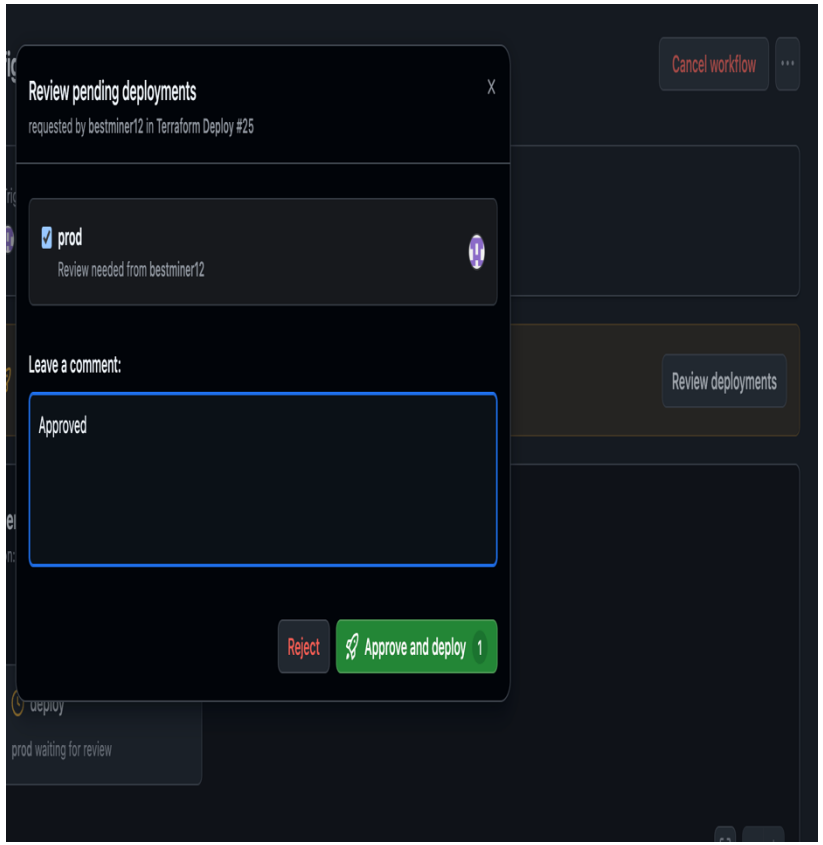


Terraform Apply, CI/CD

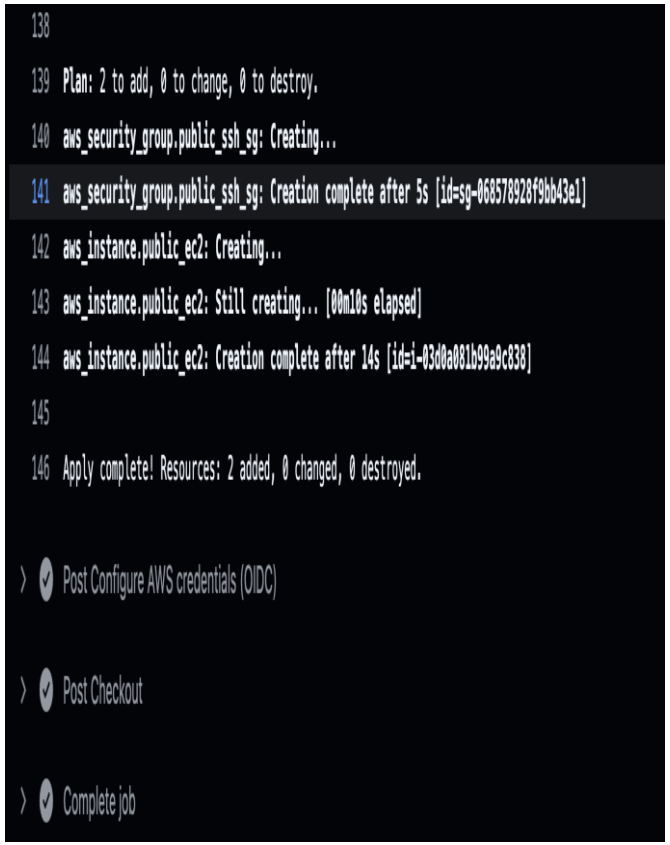
1. Terraform Code



2. Approval-based CD

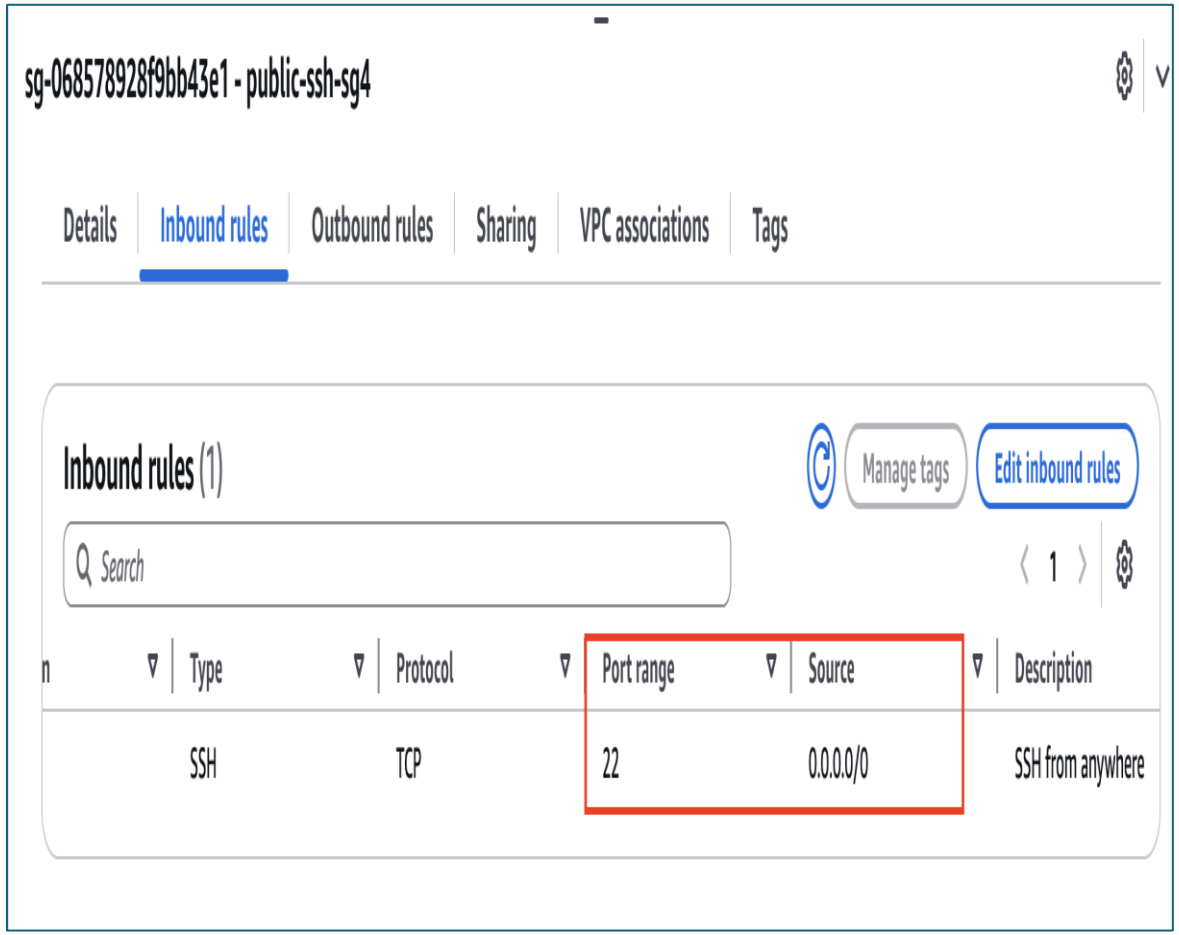


3. Resource deploy

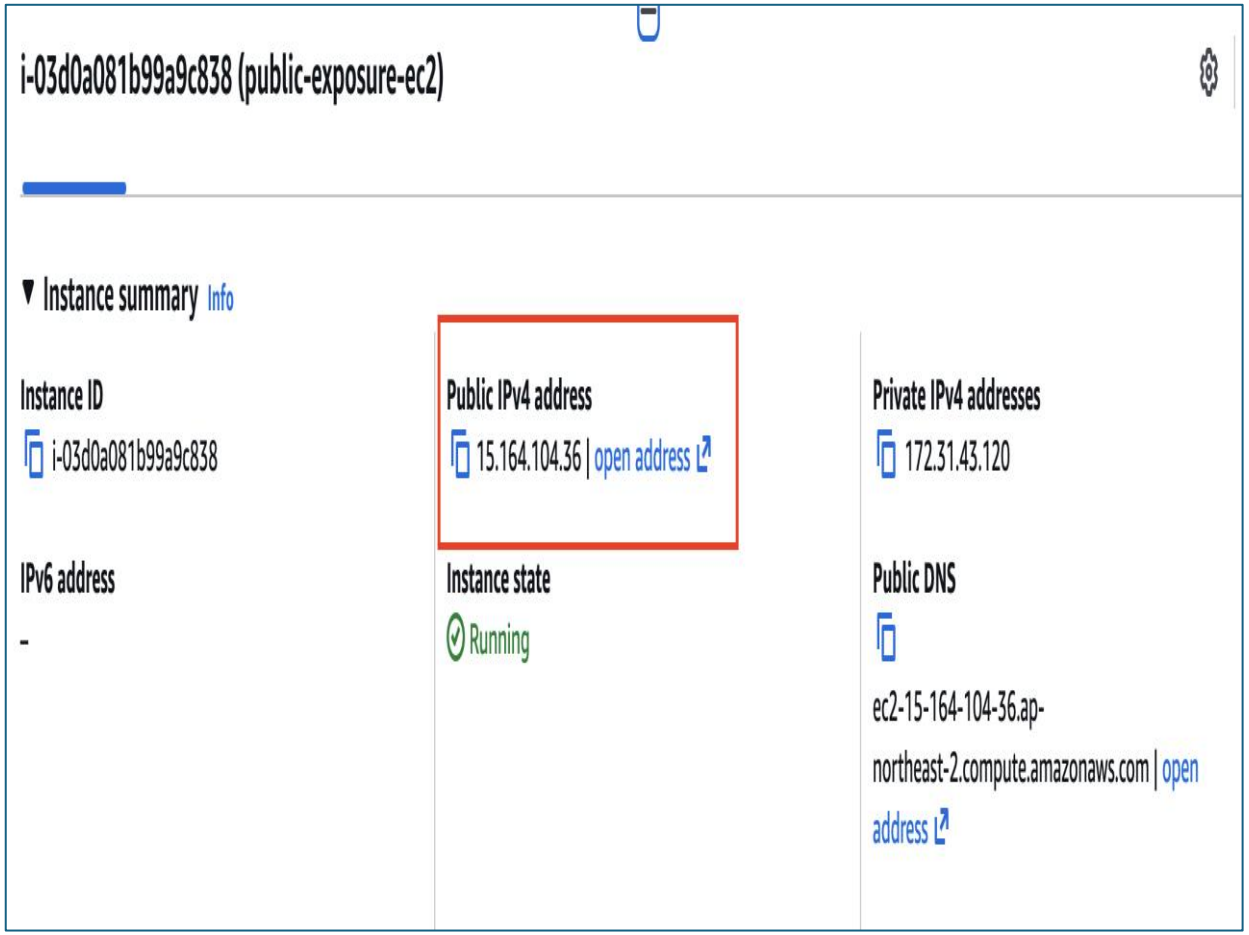


AWS Console Resource check

1. SG opend to any public ssh port

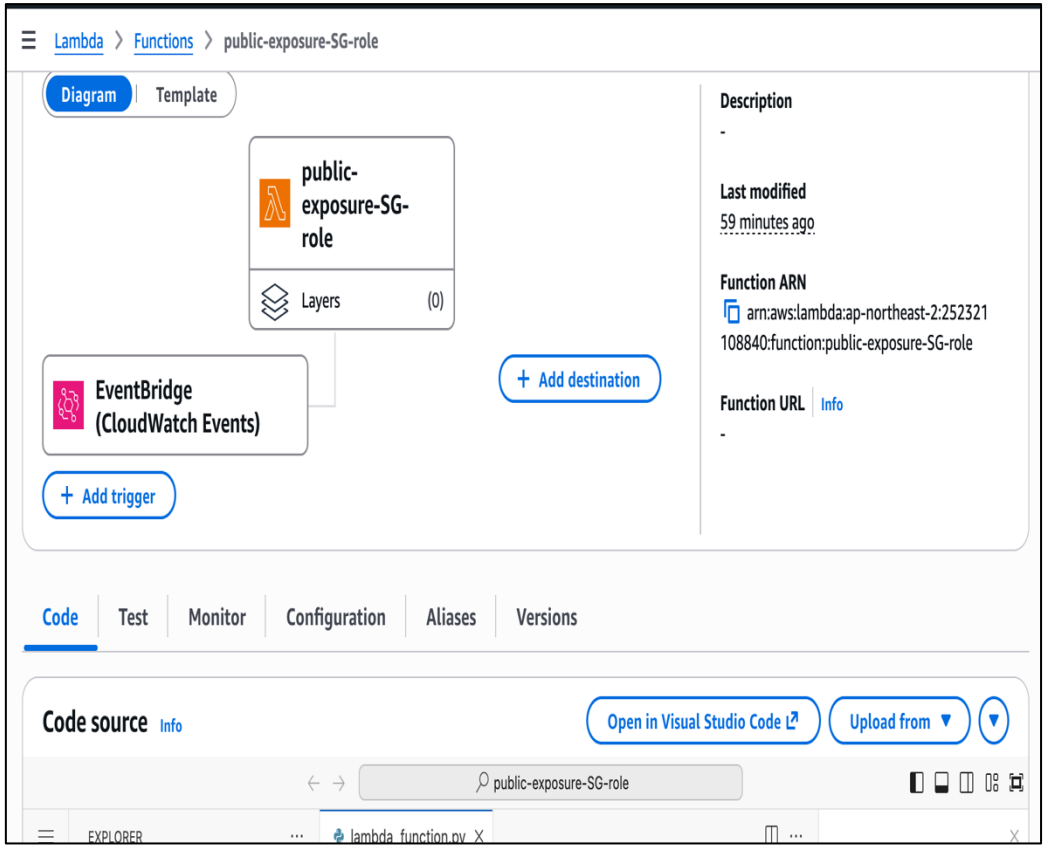


2. EC2 setting public IP

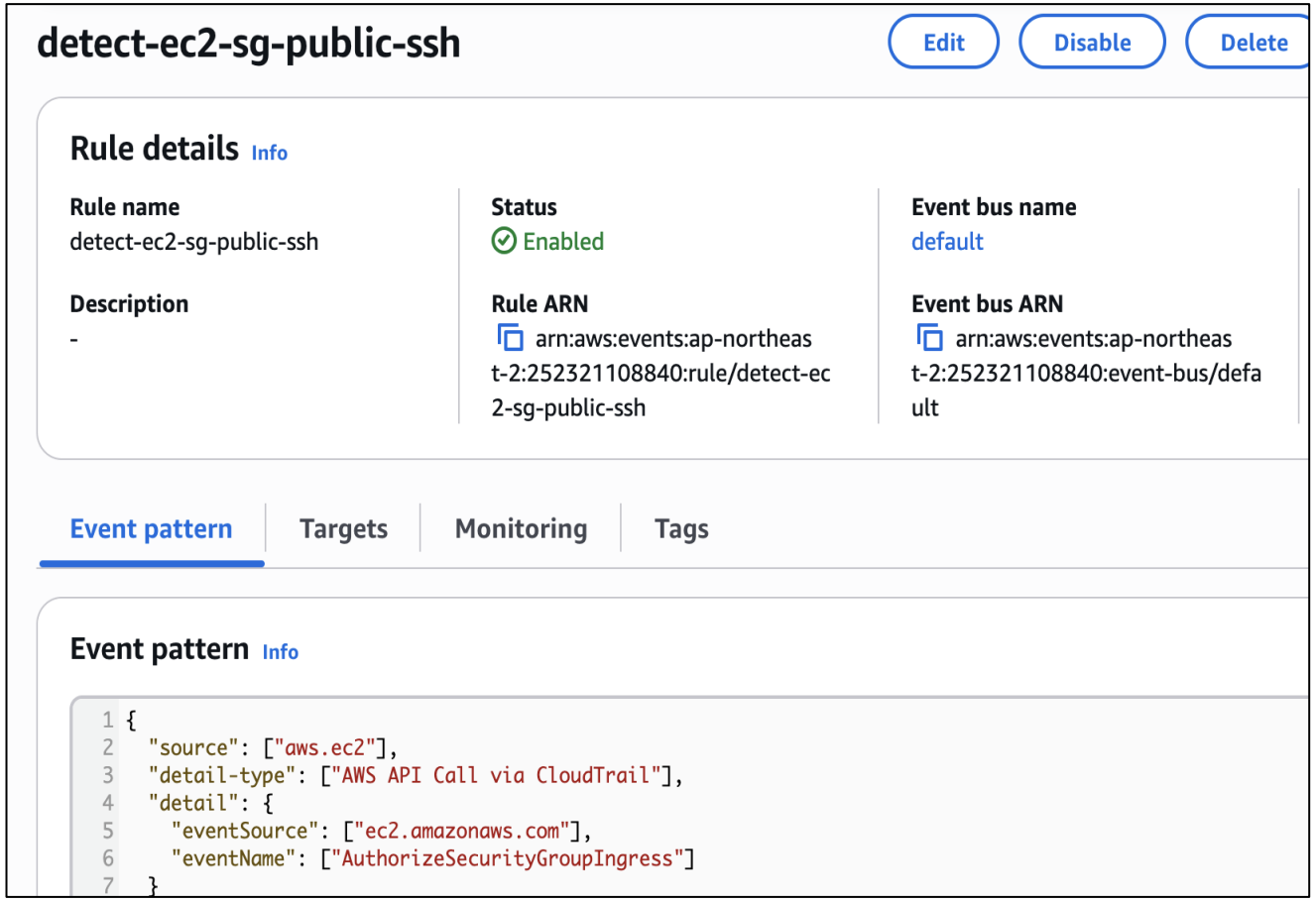


Lamda, Event bridge setting

1. Detecting Lamda setting

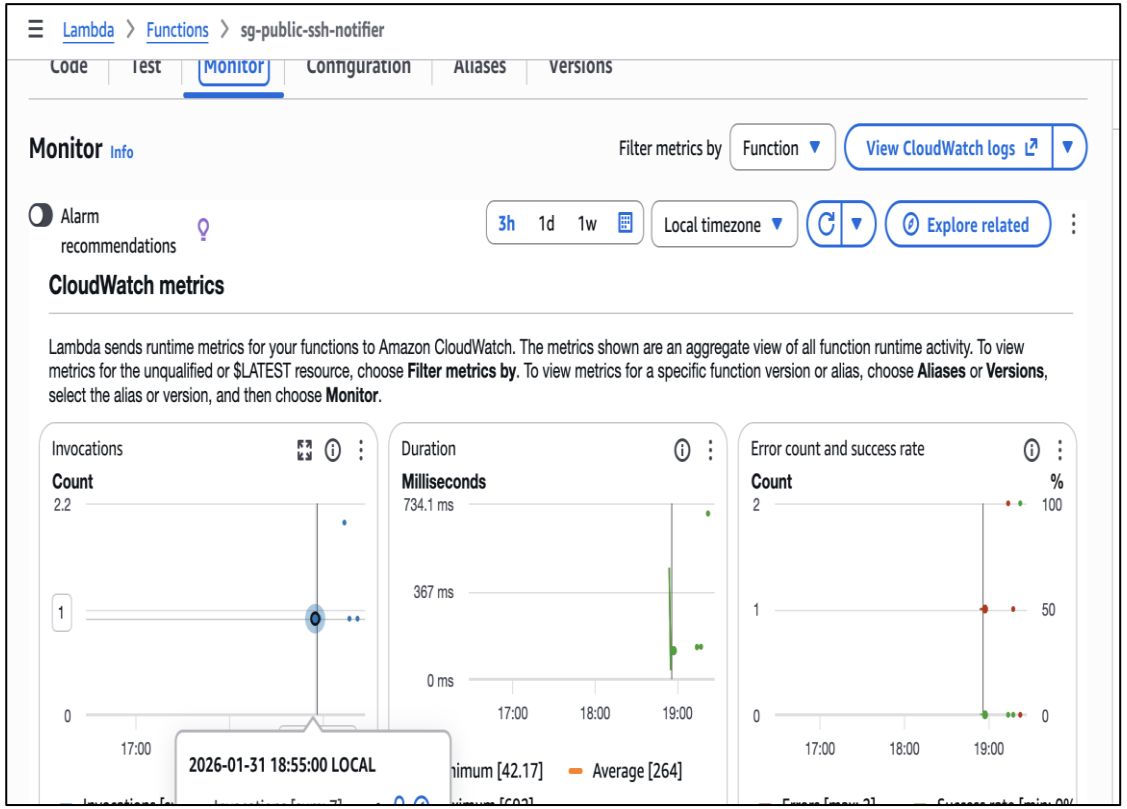


2. Event bridge setting

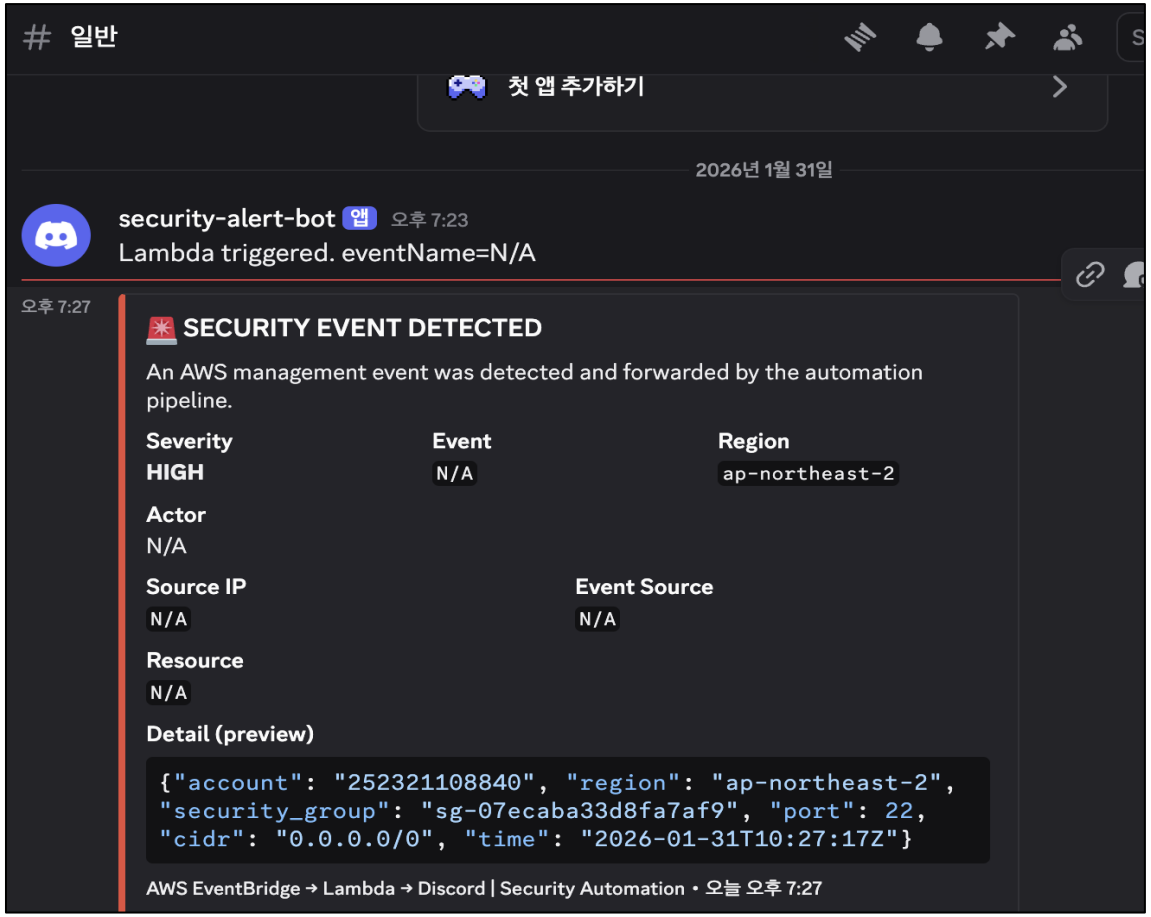


Send alert info to discord

1. Notifier lamda setting

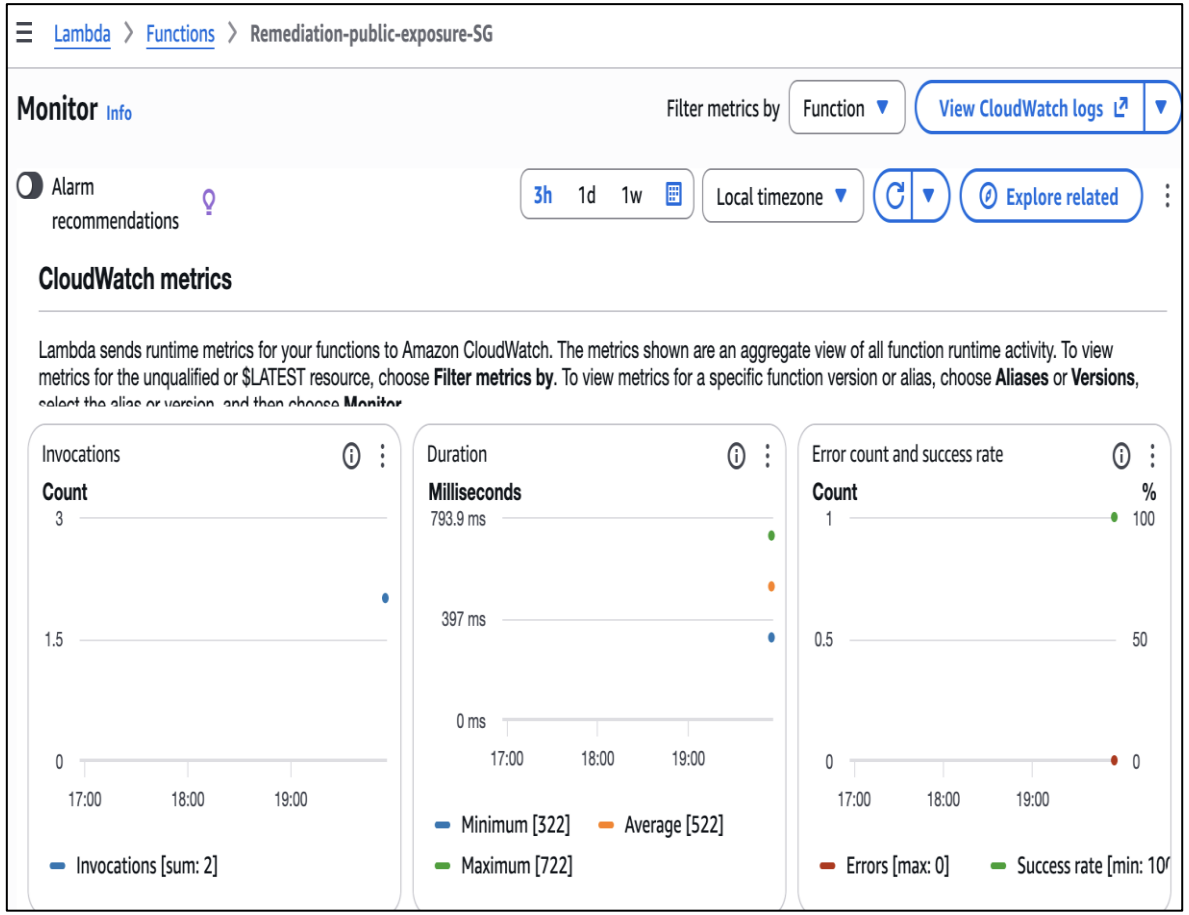
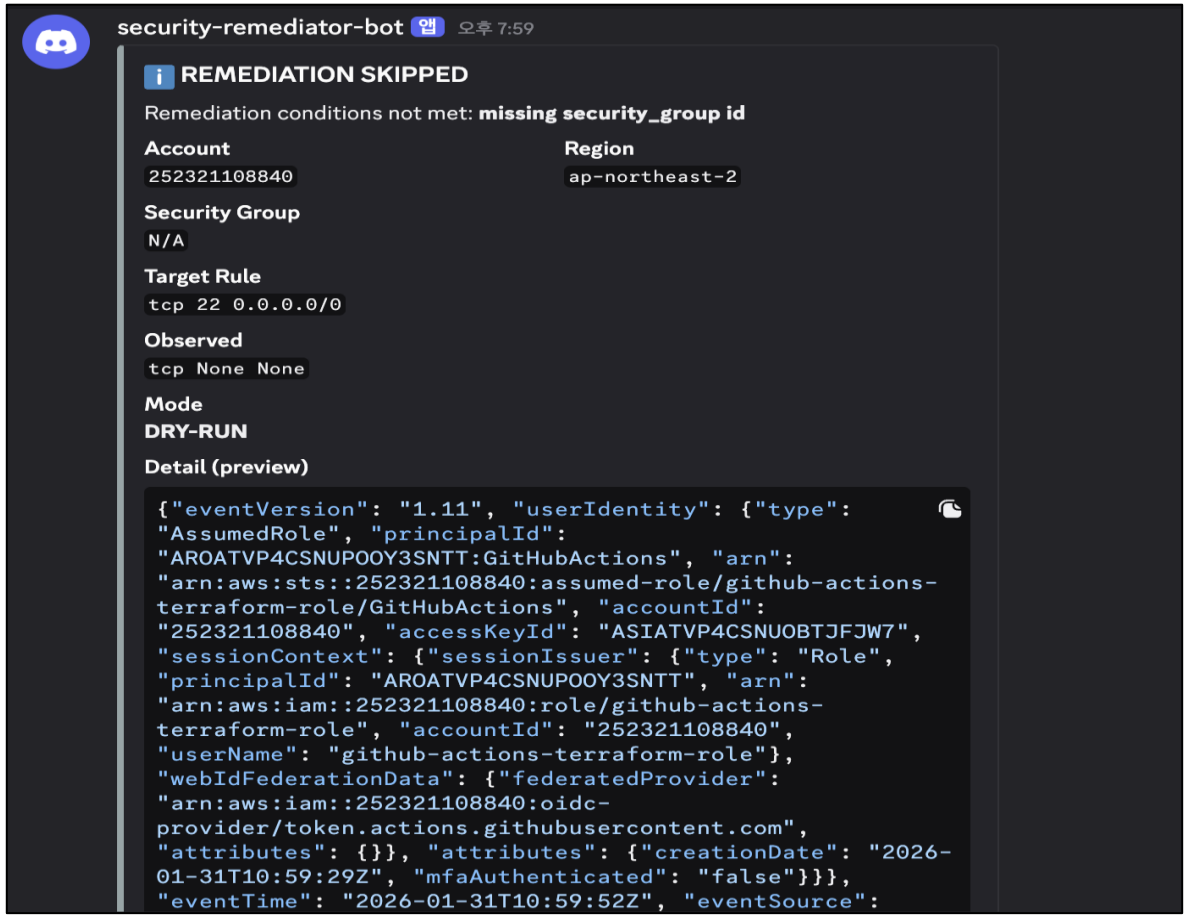


2. Send alert to discord



Automated Remediation (Security Group Hardening)

1. Remediation Lambda Configuration



2. Analyst-Driven Remediation Decision